Control Loop Hardware - Compositions

Introduction

Measuring compositions in processes are extremely important, yet tedious to measure. The downfall for composition controllers is that they're not as instantaneous as other controllers in processes like temperature and flow controllers. Composition controllers usually work interchangeably with flow controllers in getting a specific ratio from feed streams or a desired output in purity. Equipment in industry that rely on composition controllers go from continuous stirred tank reactors and distillation columns. Although measuring compositions isn't the simplest process, it remains essential in process control.

Composition Sensors & Measurements

One way in measure liquid concentrations is using a refractometer. Refractometers work by measuring the degree of how light changes in direction also know as refractive index. Usually, refractometers are independent of turbidity and viscosity, thus ensure better precision measurements [1]. Refractometers are common in measuring compositions in lab, but today in-line refractometer technology is becoming familiar in industry. Another measurement tool that's used in industry is gas and liquid chromatography, in which is the most common technique in industry. Both chromatography techniques work in similar ways as they both include an injector, columns to separate compounds, and a detector [2]. Gas chromatographs measure composition by first separating a gas into its pure gas components, and then detecting the concentration of the pure gas components [3]. Signals are sent through thermal conductivity or flame ionization detectors. This process involves first collecting a sample from a pipe, injecting the sample into a packed column of absorbent, then the detector will send the data for analysis.

Industry Importance

Composition controllers are highly important when it comes to continuous stirred tank reactors (CSTR). Their main use comes from the product stream in which the percent conversion of the reactor will also be determined once finding the composition. The product stream composition can help in determining whether more of a feed component needs to be added or regulated depending on the end composition. Also, from CSTR's end composition can play a part in safety. The importance of knowing the composition of the product stream can be to ensure no corrosive material from the feed ends up in the product. This can be damaging to equipment that may be next in process or any piping that that isn't meant to carry specific corrosive materials because it will deteriorate. This deterioration can create downtime in production, as well as a safety hazards for anyone that are around in production.

Distillation columns also rely on composition controllers in their process for purity of product. Distillate composition must be accounted for to ensure that correct purity of product is made for the consumer. Likewise, the composition of the distillate will give you an idea if the column is being ran at appropriate states, and whether parameters need to be adjusted such as the reflux and reboiler return rates. The main contribution comes from the reflux stream at the top of the column. A higher reflux back to the column ensures an increase of product purity, but also slower distillate rates. Therefore, composition controllers are important in distillation columns to have the right mix of appropriate reflux ratios to run the column in the most efficient manner.

<u>Difficulties in Measuring</u>

Measuring compositions is the most difficult parameters to measure in control processes. The largest problem with chromatography is the retention time. The retention time is the amount of time it takes for the sample to go from injection to detection in the control loop [4]. Retention time can take minutes versus other instruments in control that can take merely seconds find a reading. It also depends on the component as some components have longer retention times than others. Thus, it is unfortunate to have to wait until the composition is found to make any possible adjustments in process. Composition controls remain difficult in creating an automated environment.

<u>Control Loops</u>

A typical control loop for composition controller is showed in Figure 1. The figure shows the important of composition controllers in distillation columns as it indicates two points of where compositions are measured. In the top portion of the column, the distillate is measured to determine the amount of reflux is returned to the column by controlling the valve. As mentioned earlier, this is to ensure the appreciate purity is reached for the column's specifications. For the bottom portion of the column, the composition of the bottoms is measured to regulate the amount of steam blow in the column. The amount of steam flow is controlled by the valve which also plays a role on the bottom's composition due to the temperature of the column.

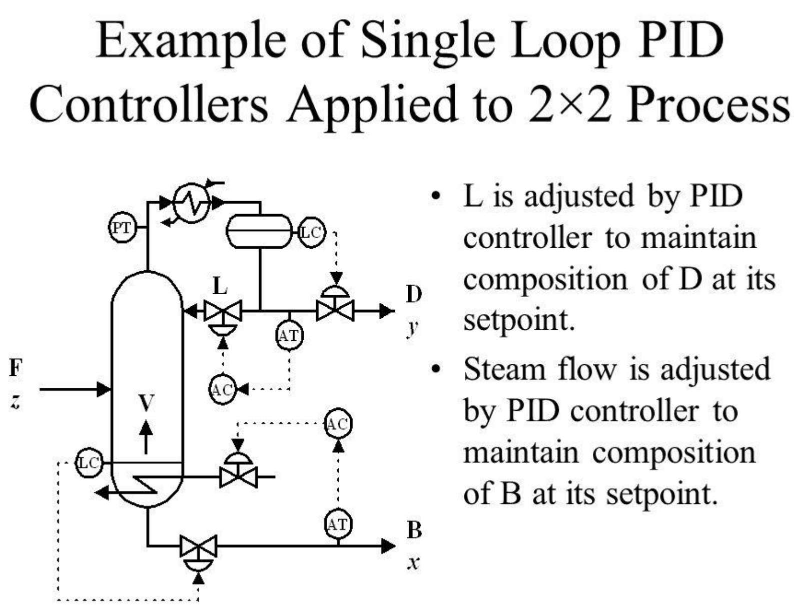

Figure 1: Composition Control Loop [5]

References

[1] “Inline Process Refractometer.” *SCHMIDT HAENSCH*, schmidt-haensch.com/products/process-instruments /in-line-process-refractometer/.

[2] Béland, D., et al. “Gas and Liquid Concentration.” *Experimental Methods and Instrumentation for Chemical Engineers (Second Edition)*, Elsevier, 17 Nov. 2017, www.sciencedirect.com/science/article/pii/B9780444637826000094.

[3] “Composition Measure.” *Composition Measure - an Overview | ScienceDirect Topics*, www.sciencedirect.com/topics/engineering/composition-measure.

[4] “Chapter 4 Chromatographic Methods.” *Chromatography*, web.njit.edu/~kebbekus/analysis /4CHROMAT.html.

[5] Controlguru. “Distillation Column Control.” *Control Guru*, controlguru.com/distillation-introduction-to-control/.